Biochemistry Cume

October 2, 2015

Total points: 100 points

1. List at least four significant similarities or differences between oxidative phosphorylation and photophosphorylation. (**12 points**)

2. The degree of reduction of each electron carrier in the respiratory chain is determined by conditions in the mitochondrion. For example, when NADH and O_2 are abundant, the steady-state degree of reduction of the carriers decreases as electrons pass from the substrate to O_2. When electron transfer is blocked, the carriers before the block become more reduced and those beyond the block become more oxidized. For each of the conditions below, predict the state of oxidation of ubiquinone and cytochromes b, c1, c, and $a_1 + a_3$. (**28 points**)
   - **A.** Abundant NADH and O_2, but cyanide is added.
   - **B.** Abundant NADH, but O_2 exhausted.
   - **C.** Abundant O_2, but NADH exhausted.
   - **D.** Abundant NADH and O_2.

3. All dehydrogenases of glycolysis and the citric acid cycle use NAD^+ (E'° for NAD^+/NADH is – 0.32 V) as electron acceptor except succinate dehydrogenase, which uses covalently bound FAD (E'° for FAD/$FADH_2$ in this enzyme is 0.05 V). Justify why FAD is more appropriate electron acceptor than NAD^+ in the dehydrogenation of succinate, based on the E'° values of fumarate/succinate (E'° = 0.03 V), NAD^+/ NADH, and the succinate dehydrogenase FAD/$FADH_2$. (Given, Faraday constant = 96.5 $kJ.V^{-1}$) (**20 points**).

4. Electron transfer translocates protons from the mitochondrial matrix to the external medium, establishing a pH gradient across the inner membrane (outside more acidic than inside). The tendency of protons to diffuse back into the matrix is the driving force for ATP synthesis by ATP synthase. During oxidative phosphorylation by a suspension of mitochondria in a medium of pH 7.4, the pH of the matrix has been measured as 7.7.
   Answer the questions below (**40 (4 x10) points**).

   - **A.** Calculate $[H^+]$ in the external medium and in the matrix under these conditions.
   - **B.** What is the outside:inside ratio of $[H^+]$ ions. Calculate the change in free energy across the mitochondrial membrane caused by the difference in $[H^+]$ concentration inside and outside the mitochondria.
   - **C.** Calculate the number of protons in a respiring liver mitochondrion, assuming its inner matrix compartment is a sphere of diameter 1.4 μm.
   - **D.** If the $\Delta G'^\circ$ needed to generate 1 mole of ATP is 30 kJ $mole^{-1}$, suggest whether the pH gradient alone is sufficiently great to generate ATP (justify your answer to receive credit).